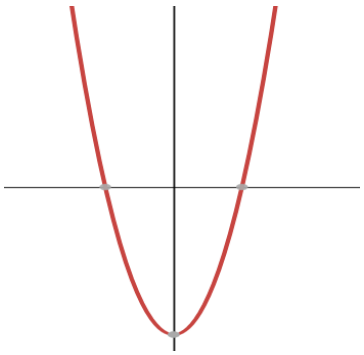
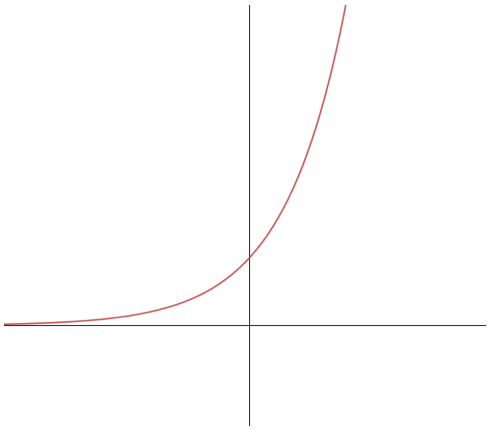
Question	Answer	Marks	Partial Marks	Guidance
8(a)(i)	6 and –6	**2**	**M1** for $x^2 = 20 + 16$ or better Or **B1** for 6 or -6	For 2 marks accept ± 6
8(a)(ii)	$\frac{7-x}{3}$ oe final answer	**2**	**M1** for $x = 7 - 3y$ or $\frac{y}{3} = \frac{7}{3} - x$ or $y - 7 = -3x$ oe or better	Oe eg $\frac{x-7}{-3}$, $\frac{x}{-3} - \frac{7}{-3}$ M1 is for switching x and y as first step or if not switching first then it is for making coefficient of $x \pm 1$ or for isolating x term.
8(a)(iii)	$9x^2 - 42x + 34$ final answer	**3**	**M1** for $(7 - 3x)^2 - 16$ [+ 1] oe **B1** for $49 - 21x - 21x + 9x^2 + k$	$(7 - 3x)^2 - 17$ with no supporting working scores M0. Allow $-42x$ for $-21x - 21x$
8(a)(iv)	Correct sketch with roots marked at –4 and 4 and y – intercept and turning point at $y = -16$ 	**4**	**B1** for correct parabola shape **B2** for roots at –4 and 4 on graph and no extras or **B1** for $(x - 4)(x + 4)$ [= 0] or for one correct root on graph or for -4 and 4 seen **B1** for turning point at (0, –16)	Condone broken line for curve For 4 marks, curve must be in all 4 quadrants with correct curvature including at ends For part marks, mark intention. **Max of 3 marks.** Condone eg slight incorrect curvature at ends, graph only in two quadrants. (More lenient than shape required for 4 marks) (0.-16) in working alone is not sufficient as it must be a turning point on graph. **See AG**

Question	Answer	Marks	Partial Marks	Guidance
8(a)(v)	[y =] $-6x - 25$	**5**	**M1** for derivative = $2x$ **M1** for x = -3 substituted into *their* derivative **B1** for (– 3, –7) soi **M1** substitution of (– 3, *their* –7) into y = *their* $-6x + c$ oe dep on 2nd M1	For M1 isw and condone poor notation eg y = $2x$ -6. For M1, condone poor notation eg y = -6 Answer $-6x + c$, c algebraic or any value implies at least M1M1 Eg implied by x = -3, y = -7 or $y = (-3)^2 - 16 = -7$ Oe eg y – *their* -7 = *their* -6(x – -3) *Their* -7 must be from using x = -3
8(b)(i)	Correct sketch with y – intercept above x – axis 	**2**	**B1** for correct shape	For 2 marks, curve should not cross the x – axis (but condone 'run along'/touching the x axis) Must tend towards x axis If intercept is given it must be 1 For B1 mark intention. Condone slight incorrect curvature eg running parallel to x axis or slight rise at ends after running parallel, graph must cross y axis. Ignore value of intercept if given. Must not cross the x axis. **See AG**
8(b)(ii)	$y = 0$	**1**		Do not accept x axis